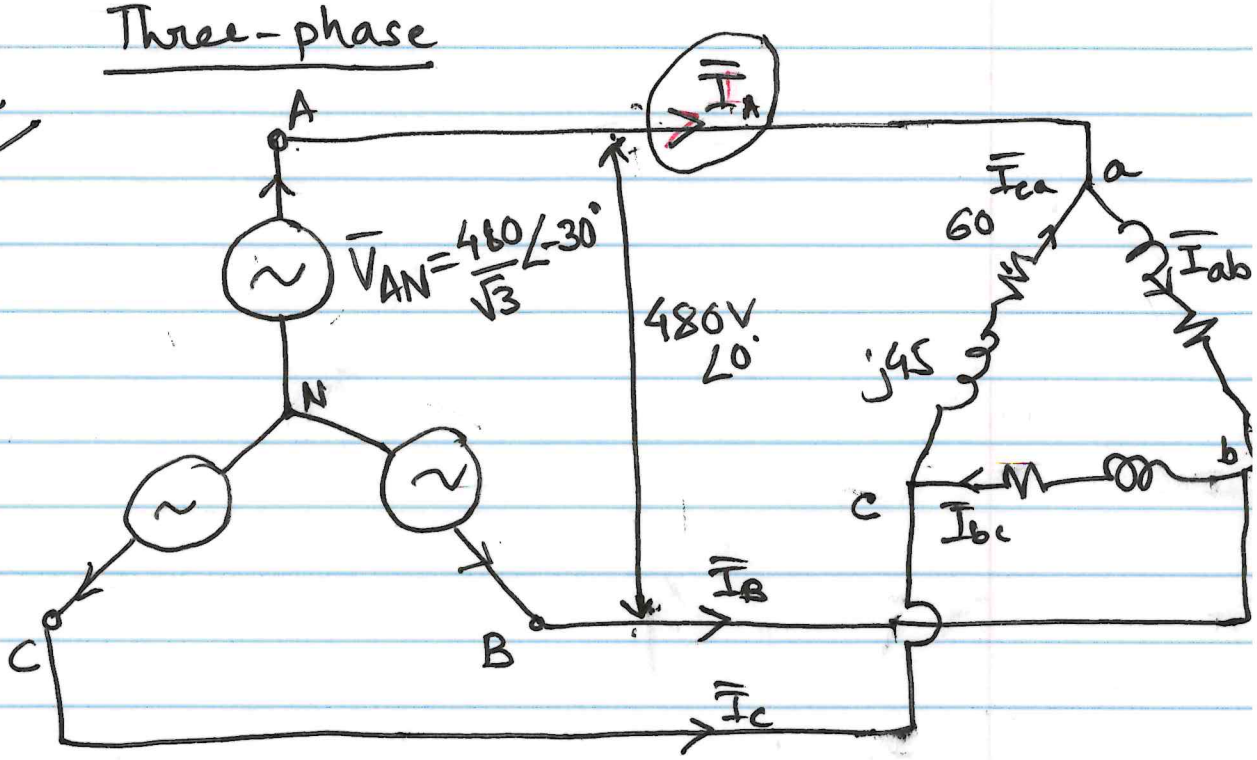
Lecture 11 (Review)

Magnetic circuit problem (coil / Loop)

Three-phase



$$Z_{delta} = 60 + j45\ \Omega$$

$$\bar{V}_{AB} = 480\angle 0°\ V \text{ (reference)}$$

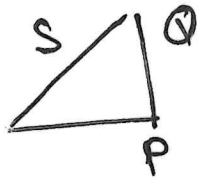
$$\bar{V}_{BC} = 480\angle -120°\ V$$

$$\bar{V}_{CA} = 480\angle -240°\ V$$

$$\bar{I}_{ab} = \frac{\bar{V}_{ab}}{Z_\Delta} = \frac{480\angle 0°}{60+j45} = 6.4\angle -36.87°\ A$$

$$\bar{I}_{bc} = \frac{\bar{V}_{bc}}{Z_\Delta} = \frac{480\angle -120°}{60+j45} = 6.4\angle -156.87°\ A$$

$$\bar{I}_{ca} = \frac{\bar{V}_{ca}}{Z_\Delta}$$



$S\cos\phi = P \quad S = \bar{V}\bar{I}^*$

$S\sin\phi = Q$

$P_{perp} + jQ_{perp} = S_{per\ phase} = \bar{V}_{ab}\bar{I}_{ab}^* = 480\angle 0^\circ \cdot 6.4\angle 36.87^\circ$ VA

$P_{total} + jQ_{total} = S_{total} = 3\,\bar{V}_{ab}\bar{I}_{ab}^*$

$= 3 \cdot 480 \cdot 6.4\angle 36.87^\circ$ VA

$= 9216\angle 36.87^\circ$ VA

Total Apparent power $|S|$

$= 9216$ VA. $\quad P + jQ = S$

Total $P = 7372.8$ W

Total $Q = 5529.6$ VAR

Alternate Path.

$P = \sqrt{3}\,|\bar{V}||\bar{I}|\cos\phi$

$P_{total} = 3 \cdot \bar{V}_{AN}\bar{I}_{AN}\cos\angle_{\bar{I}_{AN}}^{\bar{V}_{AN}}$

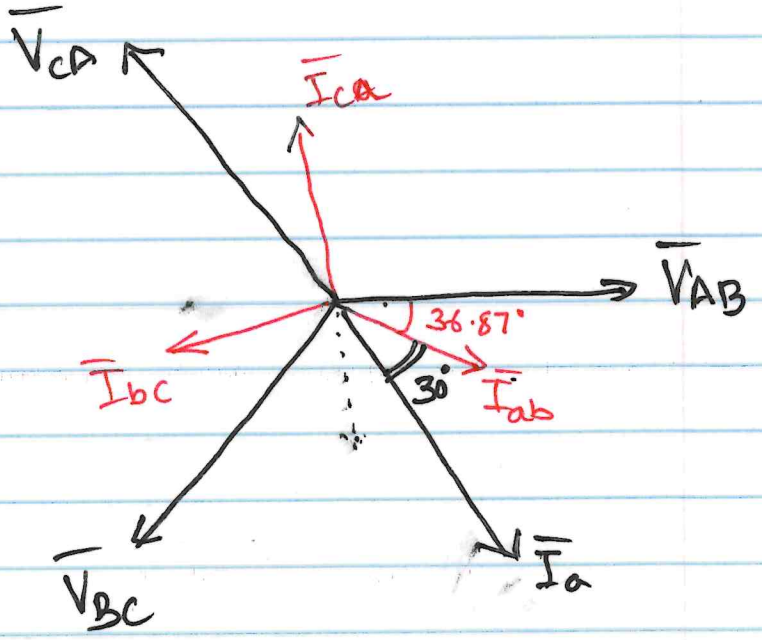
$= 3 \cdot \frac{\bar{V}_{AB}\angle -30^\circ}{\sqrt{3}} \cdot \bar{I}_{AN}\cos\angle_{\bar{I}_{AN}}^{\bar{V}_{AN}}$

KCL @ Node a.

$$\bar{I}_A = \bar{I}_{ab} - \bar{I}_{ca}$$

$$= \left(6.4\angle -36.87^\circ - 6.4\angle -276.87^\circ\right) A$$

$$\bar{I}_A = 11.09\angle -66.87^\circ \text{ A}.$$



P, Q, S, pf of original load.

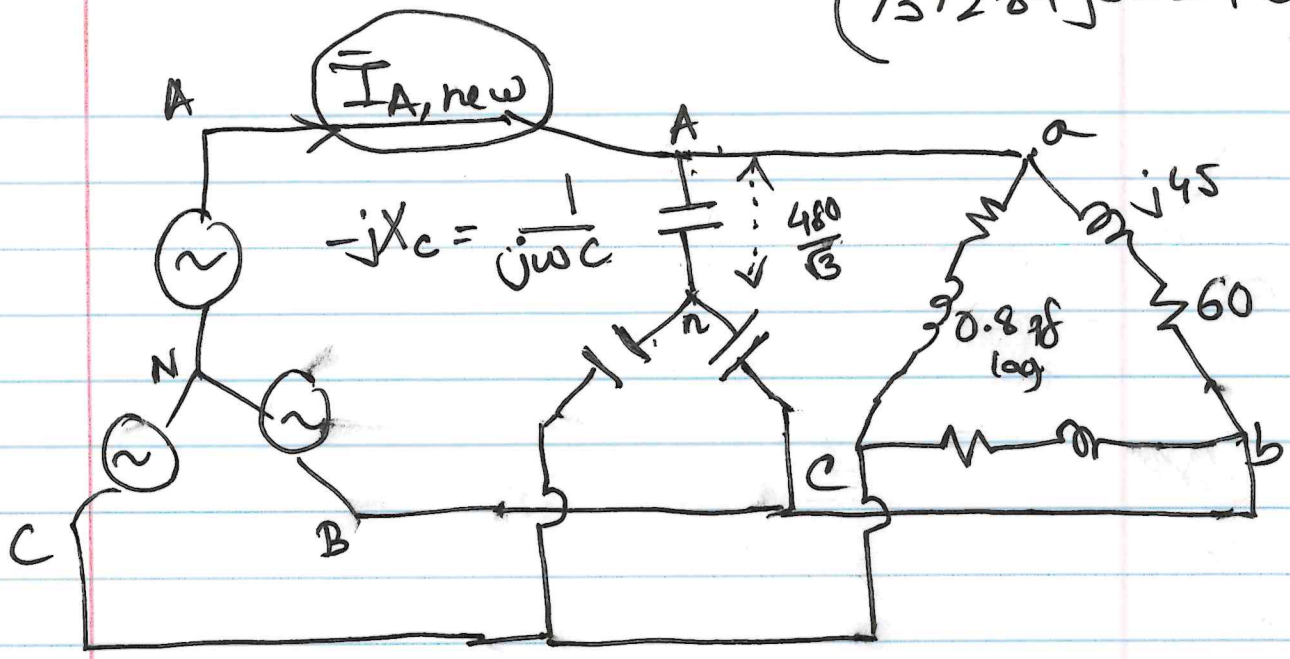
(√3)

$$P = |\bar{V}||\bar{I}|\cos\phi$$

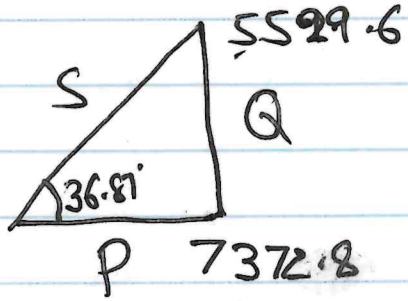
$$Q = |\bar{V}||\bar{I}|\sin\phi$$

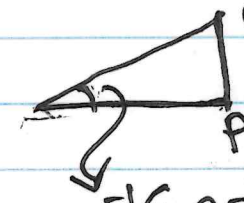
$$\bar{S} = \bar{V}\bar{I}^* \rightarrow P + jQ.$$

$(7372.8 + j5529.6)$



0.8 pf (lag) $\longrightarrow$ 0.95 pf (lag).





$\frac{Q_{new}}{P} = \tan 18^\circ$

$\cos^{-1}(0.95) = 18^\circ$ $\quad Q_{new} = 2423.3$ VAR

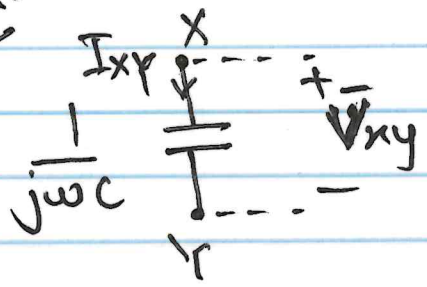
$Q_{cap} = (5529.6 - 2423.3)$ VAR

$= 3106$ VAR

$$|\bar{V}_{An}| = \frac{480}{\sqrt{3}} V$$

$$Q_{cap/ph} = \frac{3106}{3} \text{ VAR}$$

Detour:



$$\bar{V}_{xy} = \frac{1}{j\omega C} \bar{I}_{xy}$$

$$P + jQ = \bar{V}_{xy} \cdot \bar{I}_{xy}^{*}$$

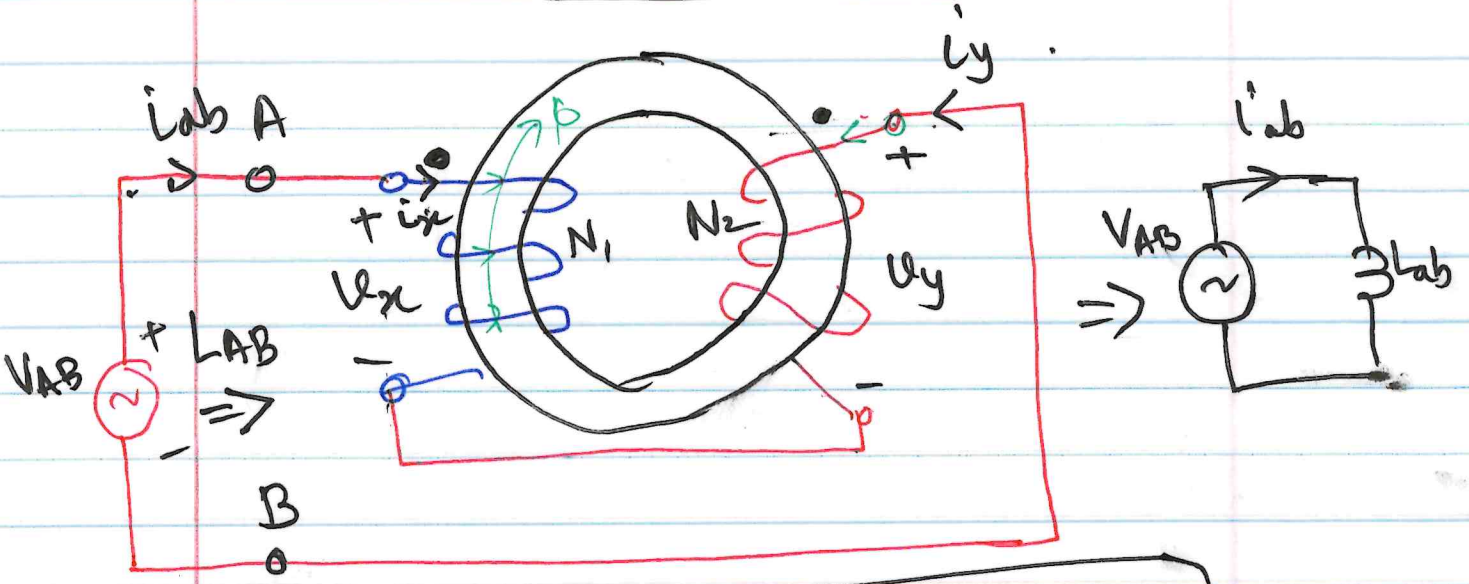
$$= \bar{V}_{xy} \cdot \left[\bar{V}_{xy} \cdot j\omega C\right]^{*}$$

$$= |\bar{V}_{xy}|^2 (-j\omega C)$$

$$\boxed{Q = |\bar{V}_{xy}|^2 \omega C}$$

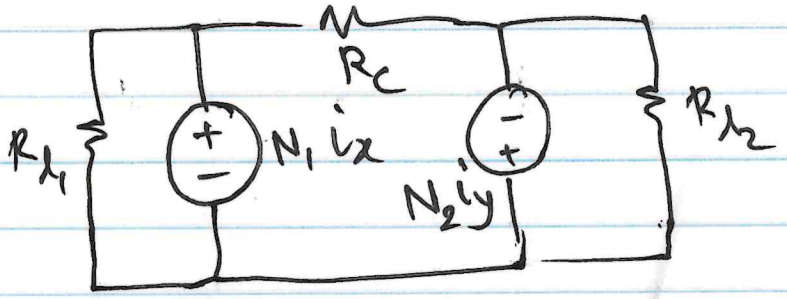
$$i_{ab} = i_x = -i_y$$

$$V_{AB} = v_x - v_y$$



$$v_x = L_{11}\frac{di_x}{dt} + M\frac{di_y}{dt}$$

$$v_y = M\frac{di_x}{dt} + L_{22}\frac{di_y}{dt}$$



$$v_{ab} = L_{AB}\cdot\frac{di'_{ab}}{dt}$$

$$V_{AB} = \left(L_{11}\frac{di_x}{dt} + M\frac{di_y}{dt} \right) - \left(M\frac{di_x}{dt} + L_{22}\frac{di_y}{dt} \right)$$

($i_x \to i_{ab}$, $i_y \to -i_{ab}$; $i_x \to i_{ab}$, $i_y \to -i_{ab}$.)

$$= \left(L_{11} - M - M + L_{22} \right)\frac{di_{ab}}{dt}$$

$$L_{11} + L_{22} - 2M$$

